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Single Body Multi-mode Mechanical Resonator Accelerometer

Abstract

A single body multi-mode mechanical resonator includes two or more spring-mass devices, one or more anchor springs, a phase 1 parallel plate electrode and a phase 2 parallel plate electrode, a plurality of interdigitated shaped combs, a plurality of interdigitated drive combs, and a plurality of interdigitated sense combs. The spring-mass devices further include two or more lumped masses and one or more anchor springs. The coupling springs couple the motion of the two or more lumped masses. The parallel plate electrodes apply forces to the in-phase mode that yield equal in magnitude but opposite in sign frequency shifts of the anti-phase mode. The shaped combs compensate for a frequency drift of the anti-phase mode and cancels first-order displacement effects. The drive combs are aligned to the anti-phase mode, with a frequency selective excitation. The sense combs are aligned to the anti-phase mode and capable of cancelling common-mode noise effects.

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Background/Summary

BACKGROUND

[0002] Inertial navigation provides a means to navigate independent of global positioning satellites (GPS) and is immune to issues related to the use of a GPS. Inertial navigation is based on fusing measurements from accelerometers and gyroscopes to estimate position, velocity, and attitude. In order to extract the actual acceleration of a vehicle while moving, a model for local gravity is needed to compensate the output of the accelerometers. In order to separate acceleration due to motion and acceleration due to gravity, a gravity gradiometer can be used. The gravity gradiometer measures the gravity gradient by using pairs of accelerometers to take differential measurements of acceleration.

Description

DESCRIPTION OF THE DRAWINGS

[0003] Features and advantages of examples of the present disclosure will be apparent by reference to the following detailed description and drawings, in which like reference numerals correspond to similar, but in some instances, not identical, components. Reference numerals or features having a previously described function may or may not be described in connection with other drawings in which they appear.

[0004] FIG. 1 is 2D schematic an example of a multi-mode mechanical resonator in an in-phase mode;

[0005] FIG. 2 is 2D schematic an example of a multi-mode mechanical resonator in an anti-phase mode;

[0006] FIG. 3 is an example of a method for real-time frequency modulated bias calibration of short-term drift in an accelerometer;

[0007] FIG. 4 is another example of a method for real-time frequency modulated bias calibration of long-term drift in an accelerometer;

[0008] FIG. 5 is another example of a method for real-time frequency modulated bias calibration of short-term drift and long-term drift in an accelerometer;

[0009] FIG. 6 is a diagram of an example of a design of an experimental multi-mode mechanical resonator;

[0010] FIG. 7A-7B are graphs of accelerometer measurement data and Allen deviation data of the accelerometer data, respectively;

[0011] FIG. 8A-8B are a scatter plot of the acceleration output and the control voltage used to maintain a constant frequency and a graph of an Allen deviation of the accelerometer output corrected using the frequency control voltage as a reference, respectively; and

[0012] FIG. 9A-9C are graphs of the raw accelerometer output vs. time, the power spectral density of the accelerometer output vs. the frequency, and Allen deviation of the accelerometer output vs. Tau (sec), respectively.

DETAILED DESCRIPTION

[0013] Gravity gradiometers are currently used to measure both acceleration due to motion and

acceleration due to gravity. An example is the Bell gradiometer. One of the core features of the Bell gradiometer is the accelerometers are mounted on a rotating platform to measure and control slow drift associated with the accelerometers. This feature helps to correct for temperature effects or the slow aging processes that influence the drift of the accelerometers. In general, gravity gradiometers are too large for some platforms. There is no suitable replacement for smaller platforms. In addition, current gravity gradiometers are expensive. Therefore, attempting to modify the known gravity gradiometers to fit smaller platforms would be inefficient and unfeasible.

[0014] In the device herein, a single-body multi-mode mechanical resonator accelerometer is used to measure both acceleration due to motion and acceleration due to gravity. The single-body multi-mode mechanical resonator accelerometer measures acceleration by detecting a variation in the frequency of an oscillator (i.e., a single body multi-mode resonator). Since the single-body multi-mode mechanical resonator accelerometer is effectively an oscillator with intentional sensitivity to acceleration, the bias drift is caused by the same effects as frequency drift in oscillators. To mitigate drift associated, for example, with temperature, a pair of multi-mode mechanical resonator accelerometers, each with only one single body multi-mode resonator, can be mounted such that their sensitive axes are aligned in opposite directions. In addition, the multi-mode mechanical resonator accelerometer is smaller and cheaper when compared to the current gravity gradiometers. Therefore, the single-body multi-mode mechanical resonator accelerometer can be efficiently incorporated into small platforms or large platforms depending on the application.

[0015] The single-body multi-mode mechanical resonator herein includes two or more spring-mass devices, one or more coupling springs, a phase 1 parallel plate electrode and a phase 2 parallel plate electrode, a plurality of interdigitated shaped combs, a plurality of interdigitated drive combs, and a plurality of interdigitated sense combs. The two or more spring-mass devices further include two or more lumped masses and one or more anchor springs. The one or more coupling springs are attached to each lumped mass and couple the motion of the two or more lumped masses. The phase 1 parallel plate electrode and the phase 2 parallel plate electrode apply forces to the in-phase mode that yield, in the presence of acceleration, equal in magnitude but opposite in sign frequency shifts of the anti-phase mode. The plurality of interdigitated shaped combs compensate for a frequency drift of the anti-phase mode and cancels first-order displacement effects of the plurality of interdigitated shaped combs. The plurality of interdigitated drive combs are aligned to resonate the anti-phase mode that has a frequency selective excitation. The plurality of interdigitated sense combs are aligned to the anti-phase mode and capable of cancelling common-mode noise effects.

[0016] Referring now to FIG. 1, a 2D schematic an example of a single body multi-mode mechanical resonator **100** in an in-phase mode is shown. FIG. 1 is for illustrative purposes only to aid in viewing and should not be construed as being limiting or directed to a particular material or materials. The single body multi-mode mechanical resonator **100** includes two or more spring-mass devices **102**. The each spring-mass device **102** includes at least one lumped mass and one or more anchor springs **106**. The example in FIG. 1 shows a single body multi-mode mechanical resonator **100** having eight anchor springs **106** total with four anchor springs **106** attached to each lumped mass of the spring mass device **102**. In another example, one or more anchor springs **106** may be present in the single body multi-mode mechanical resonator **100**. The anchor springs **106** attach the two or more lumped masses to a fixed substrate. In some examples, the fixed substrate may be any fixed substrate capable of being mounted to a chip carrier. An example of the fixed substrate is a silicon fixed substrate. The two or more spring mass devices **102** also have an in-phase mode and an anti-phase mode. This is discussed in greater detail below.

[0017] The spring-mass device **102** oscillates when the single body multi-mode mechanical resonator **100** is in use. The spring-mass devices **102** have a motion described with equations (I) and (II):

$$[00001] M_1 (\ddot{x}_1 - \ddot{x}_{1\text{Accel}}) = -K_1 (x_1 - 0) - K_c (x_1 - x_2) - F_{1A} + F_{1B} \quad (I)$$

$$M_2(\ddot{x}_2 - \ddot{x}_{2Accel}) = -K_2(x_2 - 0) - K_c(x_2 - x_1) - F_{2A} + F_{2B} \quad (II)$$

where M.sub.1 is a lumped mass of a first mass, M.sub.2 is a lumped mass of a second mass, K.sub.1 is a first anchor spring stiffness, K.sub.2 is a second anchor spring stiffness, K.sub.c is a coupling spring stiffness, x.sub.1 is a displacement of the first mass, x.sub.2 is a displacement of the second mass, $\ddot{x}$.sub.1 is an acceleration of the first mass, $\ddot{x}$.sub.2 is an acceleration of the second mass, $\ddot{x}$.sub.1Accel is a proper acceleration of the first mass, $\ddot{x}$.sub.2Accel is a proper acceleration of the second mass, F.sub.1A is a phase 1 electrostatic force applied to the first mass, F.sub.1B a phase 2 electrostatic force applied to the second mass, F.sub.2A is a phase 1 electrostatic force applied to the second mass, and F.sub.2B is a phase 2 electrostatic force applied to the first mass. The spring-mass devices **102** have two vibrational modes: an in-phase mode and an anti-phase mode. The in-phase mode (shown in FIG. **1**) is where x.sub.1=x.sub.2 in equations (I) and (II). The in-phase mode is sensitive to acceleration. Displacement of the in-phase mode due to acceleration gives rise to a frequency shift of the anti-phase mode. In an example, a frequency shift is designed to occur in a specific, known, or deterministic way. Based on the design of the single body multi-mode mechanical resonator **100** and the voltages applied to the phase 1 parallel plate electrode **108** and the phase 2 parallel plate electrode **110** (discussed in detail below), a deterministic shift in frequency occurs for a given acceleration. Once this relationship is determined using calibrated acceleration inputs, acceleration can be estimated.

[0018] An example of the single body multi-mode mechanical resonator **100** operating in an anti-phase mode is shown in FIG. **2**. The single body multi-mode mechanical resonator **100** in FIG. **2** is the same multi-mode resonator in FIG. **1**, but the single body multi-mode mechanical resonator **100** is operating in an anti-phase mode. FIG. **2** is for illustrative purposes only to aid in viewing and should not be construed as being limiting or directed to a particular material or materials. The anti-phase mode is where x.sub.1=-x.sub.2 in equations (I) and (II). The anti-phase mode is operated in a closed-loop fashion that is self-excited.

[0019] Referring back to FIG. **1**, the single body multi-mode mechanical resonator **100** includes one or more coupling springs **104**. One or more coupling springs **104** are attached to each lumped mass as shown in FIG. **1** and FIG. **2**. The coupling springs **104** couple a motion of the two or more lumped masses.

[0020] Referring back to FIG. **1**, the single body multi-mode mechanical resonator **100** includes a phase 1 parallel plate electrode **108** and a phase 2 parallel plate electrode **110**. The phase 1 parallel plate electrode **108** and the phase 2 parallel plate electrode **110** are attached to each lumped mass (i.e., each lumped mass has a phase 1 parallel plate electrode **108** and a phase 2 parallel plate electrode **110**) as shown in FIG. **1** and FIG. **1**. The phase 1 parallel plate electrode **108** and the phase 2 parallel plate electrode **110** apply forces to the in-phase mode that, in the presence of acceleration, yield equal in magnitude, but opposite in sign frequency shifts of the anti-phase mode. The anti-phase mode (i.e., f-mode) is insensitive to external acceleration and the in-phase mode (i.e., a-mode) is sensitive to acceleration. Anti-phase mode is operated in a positive feedback loop (output of the single body multi-mode mechanical resonator **100** is connected to input with a positive gain) such that electrodes **108**, **110** amplitude is kept at a constant value. Displacement of the in-phase mode has the ability to modulate the stiffness of the anti-phase mode, and in turn, the frequency, of the anti-phase mode when operated as a single body multi-mode mechanical resonator via an electrostatic effect associated with the phase 1 and phase 2 parallel plate electrodes **108**, **110**. The stiffness of the anti-phase mode is derived from the anchor springs **106**, the coupling springs **104**, and the electrodes **108**, **110**. As the multi-mode resonator gets closer to the electrodes **108**, **110** the frequency decreases. The opposite effect occurs as the multi-mode resonator moves away from the electrodes **108**, **110**, the frequency increases. When the electrodes **108**, **110** are aligned in in-phase mode (i.e., each electrode **108**, **110** is active), for a given acceleration input, each give rise to equal in magnitude, but opposite in sign frequency shifts.

[0021] Referring back to FIG. 1, the single body multi-mode mechanical resonator **100** further includes a plurality of interdigitated shaped combs **112**. The plurality of interdigitated shaped combs **112** is attached to each lumped mass as shown in FIG. 1 and FIG. 2. The plurality of interdigitated shaped combs **112** compensates for a frequency drift of the drive mode and cancels first-order displacement effects of the plurality of interdigitated shaped combs. In an example, frequency drift refers to undesirable effects, such as a change in frequency due to temperature variations. The profile of the plurality of interdigitated shaped combs **112** is designed such that the force produced is linearly related to the displacement. The plurality of interdigitated shaped combs **112** have a stationary finger and a moving finger that produce the force. The force between the stationary finger and the moving finger can be represented by the following equation (III):

$$[00002] F(x) = \epsilon_0 t V^2 / h(x) \quad (III)$$

where $F(x)$ the amount of force generated between a pair of a stationary finger and a moving finger, $\epsilon_{\text{sub}.0}$ is the permittivity of free space, t is the thickness of the stationary finger and a moving finger, V is the voltage difference between the stationary finger and moving finger, and $h(x)$ is size of the gap between a stationary finger and a moving finger.

[0022] To generate a specific force ($F(x)$) between a pair of stationary and moving fingers, a gap between the stationary finger and the moving finger needs to be a specific shape or profile. In an example, one of the fingers is selected to have rectangular geometry and the other has a curved profile to generate the gap. Thus, to produce a force that is linear with respect to x , the size of the gap between the stationary finger and moving finger is represented by the following equation (IV):

$$[00003] h(x) = g_{\text{sub}.0} / (1 + x / x_{\text{CL}}) \quad (IV)$$

where $g_{\text{sub}.0}$ is the gap between a straight comb and the maximum distance between the straight comb and a curved comb and x_{CL} is a dimensional constant. In the equation (IV), the straight comb is the stationary finger or moving finger and the curved comb is different from the straight comb and either the stationary finger or the moving finger. In an example, the movement distance of the moving fingers can be equal to or less than 10 microns. In another example, the movement distance can be adjusted based on the needs to reach a specific displacement for low noise operating and accuracy of the fabrication methods to match the gap profile described in equation (IV). In an example, the plurality of interdigitated shaped combs **112** may be present in any amount that compensates for slow frequency drifts. In some examples, the plurality of interdigitated shaped combs **112** is present in an amount of equal to or greater than 500 sets of the stationary finger and the moving finger.

[0023] Referring back to FIG. 1, the single body multi-mode mechanical resonator **100** also includes a plurality of interdigitated drive combs **114**. The plurality of interdigitated drive combs **114** are attached to each lumped mass as shown in FIG. 1 and FIG. 2. The plurality of interdigitated drive combs **114** are aligned to the anti-phase mode that, with a frequency selective excitation, can excite the anti-phase mode with a low voltage when the single body multi-mode mechanical resonator **100** is operated within a low-pressure environment, which minimizes the effect of viscous damping. Frequency selective excitation refers to when the anti-phase mode exhibits resonance phenomena. The resonance phenomena has the largest displacement at specific resonant frequencies. Moreover, there are specific phase relationships between the input and output at these resonant frequencies such that a feedback system that compensates for the phase delay and damping can amplify oscillations at these frequencies subject to nonlinear effects. In an example, the low-pressure environment is equal to or less than 10 mTorr. In the example shown in FIG. 1, there is one set of a plurality of interdigitated drive combs **114** for each spring-mass device **102**. In other examples, there are one or more sets of interdigitated drive combs **114** for each spring-mass device **102** where each set of interdigitated drive combs **114** includes a plurality of interdigitated drive combs.

[0024] Referring now to FIG. 1, the single body multi-mode mechanical resonator **100** includes a

plurality of interdigitated sense combs **116** that are attached to each lumped mass as shown in FIG. **1** and FIG. **2**. The plurality of interdigitated sense combs **116** are aligned to the anti-phase mode, are grounded when the single body multi-mode mechanical resonator **100** is polarized, and are capable of cancelling common-mode noise effects. The motion of the anti-phase mode induces a charge in one half of the plurality of interdigitated sense combs **116** that is equal in magnitude, but opposite in sign to the charge in the other one half of plurality of interdigitated sense combs **116** such that a differential measurement can cancel common-mode noise effects. Common-mode noise effects are noise effects shared between various nodes that are independent of the orientation or relative position of these nodes. An example of a common-mode noise effect is the noise that arises due to the bias voltage applied to the single body multi-mode mechanical resonator **100**. In the example of the single body multi-mode mechanical resonator **100** in FIG. **1**, each spring-mass device **102** has four sets of interdigitated sense combs **116** where each set of interdigitated sense combs **116** includes a plurality of interdigitated sense combs **116**. In another example, each spring-mass device **102** has one or more sets of interdigitated sense combs **116** where each set of interdigitated sense combs **116** includes a plurality of interdigitated sense combs **116**.

[0025] Overall, the single body multi-mode mechanical resonator **100** may be composed of a variety of materials. For example, the single body multi-mode mechanical resonator **100** herein may be composed of silicon, quartz, n or p-type doped silicon, lead zirconate titanate, or a combination thereof. The selective deposition of metallic layers may be used for electrically connecting components to the single body multi-mode mechanical resonator **100** to an accelerometer or enhancing the operation of the single body multi-mode mechanical resonator **100**. For example, a 20 nm layer of Chromium can be evaporated as a first layer to enhance the adhesion of a thicker 500 nm layer of gold for electrical contacts. The location of these contact are based on the need to electrically connect the device to other systems and placed with rigid connections to a lower fixed substrate. This layer can be placed over a larger area, but placement of the electrical layer over areas designed to move can result in a bending or curling of the device. Chrome is used as an adhesion layer to make sure the gold sticks. In some examples, titanium may be used instead of chrome as an adhesion layer. In an example, the single body multi-mode mechanical resonator **100** may have a device layer thickness ranging from about 2 μm to about 500 μm .

[0026] The single body multi-mode mechanical resonator **100** is also capable of being integrated with electronic subsystems to create the single body multi-mode mechanical resonator accelerometer. In an example, only one single body multi-mode mechanical resonator **100** is needed in the single body multi-mode mechanical resonator accelerometer. In an example, the electronic subsystems include a substrate, a chip carrier, an interface electronics circuit board or a combination thereof. In an example, the single body multi-mode mechanical resonator **100** is mounted to a fixed substrate (e.g., a silicon substrate). The fixed substrate is mounted to a chip carrier and the chip carrier is mounted to an interface electronics circuit board. In some examples, a leveling mount may also be used where the interface electronics circuit board, chip carrier, and fixed substrate are mounted on the leveling mount. In addition, the interface electronics circuit board, chip carrier, and fixed substrate, and, in some examples, the leveling mount are in a vacuum enclosure to vacuum seal the components.

[0027] Other known components required for the use of an accelerometer are also used to form a functional accelerometer, such as a positive feedback system, an automatic gain control circuit, a bias reference or voltage source, a modulation reference, and an accelerometer controller. The accelerometer controller is able to measure or demodulate the frequency of the single body multi-mode accelerometer **100**, further process the demodulated frequency to extract information at harmonics of the modulation reference, use this data to estimate acceleration, and provide an input to a control system that adjust the voltage applied to the plurality of interdigitated shaped combs **112** such that the mean, average, or 0-th harmonic of the modulation reference is maintained at a constant value. The accelerometer controller also has the ability to store or transmit data to allow

for further processing, and implement algorithms to compensate for short or long-term drift.

[0028] Referring now to FIG. 3, an example of a method **300** for real-time frequency modulated bias calibration of short-term drift in the single-body multi-mode mechanical resonator accelerometer is shown. In the method **300**, an extracted accelerometer output is measured **302** where the extracted accelerometer output drifts over time, thereby causing an inaccurate extracted accelerometer output. The drift occurs because effects that cause the frequency of the anti-phase mode to drift, for example temperature variations, cannot be separated from acceleration from linear motion that also causes the frequency of the anti-phase mode to shift. In an example, the extracted output of the accelerometer is obtained by measuring the output of an interface electronic circuit that corresponds to a displacement of two or more spring-mass devices **102** in an anti-phase mode, a differential charge produced by motion of two or more spring-mass devices **102** in the anti-phase mode, or a combination thereof. In some examples, the measurement may be further processed to obtain the final extracted output of the accelerometer prior to using the autoregressive filter.

[0029] Referring back to FIG. 3, the method **300** further includes applying an autoregressive filter to the extracted accelerometer output **304**, thereby removing short-term noise. The autoregressive filter removes or filters stationary noise processes from the extracted accelerometer output. The autoregressive filter is applied using the following equation (V):

$$[00004] \ y_t = \delta_{\text{sub}.0} + \delta_{\text{sub}.1} t + \sum_{k=0}^p \phi_{\text{sub}.k} y_{t-k} + \epsilon_{\text{sub}.t} \quad (V)$$

where t is time, $\delta_{\text{sub}.0}$ is linear offset, $\delta_{\text{sub}.1}$ is linear trend with time, p is an autoregressive model order, $\phi_{\text{sub}.k}$ is an autoregressive model parameter, $y_{\text{sub}.t-k}$ is a model value at discrete time (t-k), k is a k-th term of the summation, and $\epsilon_{\text{sub}.t}$ is uncorrelated noise process at discrete time t. In some examples, the autoregressive filter is fit to an autoregressive model that automatically selects the model order based on an Akaike information criterion. In another example, the autoregressive filter is fit to an autoregressive model that automatically selects the model order based on an Akaike information criterion, another metric that penalizes high model orders to minimize the potential of overfitting, or a combination thereof.

[0030] Referring back to FIG. 3, the method **300** includes calculating a corrected accelerometer output **306**, thereby determining an acceleration. The method **300** may be performed by a computer processor, an application-specific integrated circuit, a field programmable gate array, or a microcontroller. Specifically, a computer processor, an application-specific integrated circuit, a field programmable gate array, or a microcontroller is used to record a measurement of the extracted output of the accelerometer, apply the autoregressive filter, and calculate the corrected accelerometer output.

[0031] Referring now to FIG. 4, an example of a method **400** for real-time frequency modulated bias calibration of long-term drift in a single-body multi-mode mechanical resonator accelerometer is shown. The method **400** includes measuring an extracted accelerometer output **402**, wherein the extracted accelerometer output drifts over time, thereby causing an inaccurate extracted accelerometer output. The method of extracting accelerometer output **402** is the same as previously described herein in method **300** for measuring an extracted accelerometer output **302**.

[0032] Referring back to FIG. 4, the method **400** includes measuring the frequency of a single-body multi-mode mechanical resonator accelerometer **404**, thereby obtaining frequency data to extract a mean value. In an example, the mean value of the frequency is exacted with a digital low-pass filter. In an example, a digital low-pass filter can be implemented with either digital electronic circuits or a software algorithm provided that when the input signal to the filter is considered a composition of many signals at various frequencies, frequencies above a specified cutoff frequency are attenuated. For the low-pass filter to extract the mean value, this cut-off frequency must below the frequency of a modulation reference frequency. The cut-off frequency may vary depending on the application of the single body multi-mode mechanical resonator accelerometer.

[0033] Referring back to FIG. 4, the method **400** includes adjusting a voltage applied to a plurality of interdigitated shaped combs **112** using the mean value and a proportional-integral-derivative controller or a proportional-integral controller (PID or PI controller) **406**, thereby forming a control voltage to maintain a constant mean frequency value. The plurality of interdigitated shaped combs **112** are the same plurality of interdigitated shaped combs **112** as previously described herein. In an example, the applied voltage is enough to change the stiffness of the single body multi-mode resonator **100** to overcome changes due to effects, such as changing temperature. For example, a range of about 1% to about 5% change in stiffness of the single body multi-mode mechanical resonator **100** to overcome changes due to the temperature associated with the choice of material (e.g. silicon) and the anticipated fluctuations in ambient temperature. In another example, the amount of voltage applied is based on the stiffness of the anti-phase mode, the number of the plurality of interdigitated shaped combs **112**, and the geometry of the plurality of interdigitated shaped combs **112**, in particular, the dimensional constant of the plurality of interdigitated shaped combs **112**. In an example, the PID controller may be any controller capable of adjusting the voltage such that the error is zero.

[0034] Referring back to FIG. 4, the method **400** includes determining an amount of drift in the accelerometer using the control voltage **408**. In an example, to determine the amount of drift in the accelerometer, an assumption is made that there is a linear relationship between the drift and the control voltage. The correlation coefficient R quantifies the strength of the linear relationship (shown in FIG. 8A). An R of either 1 or -1 is a perfect linear relationship. Equation (VI) shows how the corrected accelerometer output (A1S_corrected) is determined:

$$[00005] A1S_corrected = A1S + m * y_A_LP + b$$

Where y_A_LP is the low pass filtered version of the control voltage, A1S is the output of the accelerometer, m is the slope from the linear fit, and b is the intercept of the linear fit. These values are estimated using linear regression. Once this relationship is learned, future drift of the accelerometer can be corrected by using the fitted relationship and the current value for the control voltage. In an example, **408** can be performed by a computer processor to determine the amount of drift for the subsequent calculation below in **410**.

[0035] Referring back to FIG. 4, the method **400** includes calculating a corrected accelerometer output using the amount of drift **410**, thereby determining an acceleration. Based on the fitted linear relationship discussed above in **408**, any changes in control voltage can be assumed to produce a linear change in the output of the accelerometer. Therefore, as stated above, future drift of the accelerometer can be corrected by using the fitted relationship and the current value for the control voltage. In an example, **410** can be performed by a computer processor to determine the acceleration.

[0036] In a specific example of method **400**, the plurality of interdigitated shaped combs **112** have one end attached to the body of the single body multi-mode mechanical resonator **100** and the other end is fixed in space. A bias voltage is applied to the lumped masses of the single body multi-mode mechanical resonator **100** causing movement and producing a charge. This charge is then converted to a voltage with circuits on the interface electronics board. The voltage is then connected to an analog to digital converter (ADC) on a single board computer. The single board computer runs a program to control the accelerometer. Part of this program is a software phase locked loop (PLL) that extracts the frequency of the single body multi-mode resonator **100**. A PLL can be either an analog circuit, digital circuit or software algorithm. The key characteristic of a PLL is this control system is the adjustment of the frequency of the single body multi-mode resonator **100** internal to this device to maintain a constant phase relationship between this single body multi-mode resonator **100** and the input. When the phase between two signals is constant, both signals have the same frequency, so the frequency of the internal single body multi-mode resonator **100** can be used to track the frequency of the input signal. A second PLL is used to track the frequency and phase of the signal applied to the phase 1 parallel plate electrode **108**. Following compensation of the

extracted phase, this can be used to demodulate a given harmonic. This is done by multiplying the extracted reference by frequency, and then low-pass filtering to remove higher-order harmonics. [0037] As stated in the example, the method **400** is performed by a computer processor (i.e., a single board computer). The method **400** can also be performed by an application-specific integrated circuit, a field programmable gate array, or a microcontroller. The computer processor, application-specific integrated circuit, field programmable gate array, or microcontroller is used to record a measurement output of the extracted accelerometer output, record a frequency measurement of the single body multi-mode resonator **100**, adjust the voltage applied to the plurality of interdigitated shaped combs **112**, determine the amount of drift in the accelerometer, and calculate the corrected accelerometer output.

[0038] Referring now to FIG. 5, an example of method **500** for real-time frequency modulated bias calibration of short-term drift and long-term drift in an accelerometer is shown. The method **500** includes measuring an extracted accelerometer output **502**, wherein the extracted accelerometer output drifts over time, thereby causing an inaccurate extracted accelerometer output. The measured extracted accelerometer output **502** is the same as previously described herein in methods **300**, **400** for measuring an extracted accelerometer output **302**, **402**.

[0039] Referring back to FIG. 5, the method **500** includes performing determining a corrected short-term accelerometer output, determining a corrected long-term accelerometer output, or a combination thereof. In an example, if determining a corrected short-term accelerometer output and determining a corrected long-term accelerometer output are both performed, the determinations may be performed sequentially or simultaneously. In another example, if determining a corrected short-term accelerometer output and determining a corrected long-term accelerometer output are both performed, determining the long-term accelerometer output occurs before determining the corrected short-term accelerometer output.

[0040] When determining a corrected short-term accelerometer output is performed in method **500**, the short-term accelerometer output is determined by applying an autoregressive filter to the extracted accelerometer output **504**, thereby removing short-term noise and calculating the corrected short-term accelerometer output **506**, thereby determining a short-term acceleration. The applying an autoregressive filter to the extracted accelerometer output **504** and calculating the corrected short-term accelerometer output **506** are the same as applying an autoregressive filter to the extracted accelerometer output **304** and calculating the corrected short-term accelerometer output **306** as previously described herein for the method **300**.

[0041] When determining a corrected long-term accelerometer output is performed in method **500**, the long-term accelerometer output is determined by measuring the frequency of the single body multi-mode resonator **100508**, thereby obtaining frequency data to extract a mean value, adjusting a voltage applied to a plurality of interdigitated shaped combs **112** using the mean value and a control algorithm **510**, thereby forming a control voltage to maintain a constant mean frequency value, determining an amount of drift in the accelerometer using the control voltage **512**, and calculating the corrected long-term accelerometer output **514**, thereby determining a long-term acceleration. The measuring the frequency of the single body multi-mode resonator **100508**, adjusting a voltage applied to a plurality of interdigitated shaped combs **112** using the mean value and a control algorithm **510**, determining an amount of drift in the accelerometer using the control voltage **512**, and calculating the corrected long-term accelerometer output **514** is the same as measuring the frequency of the single body multi-mode resonator **100 404**, adjusting a voltage applied to a plurality of interdigitated shaped combs **112** using the mean value and a control algorithm **406**, determining an amount of drift in the accelerometer using the control voltage **408**, and calculating the corrected long-term accelerometer output **410** as previously described herein for method **400**.

[0042] To further illustrate the present disclosure, examples are given herein. These examples are provided for illustrative purposes and are not to be construed as limiting the scope of the present

disclosure.

EXAMPLE

[0043] For the fabrication of the single body multi-mode mechanical resonator, the SOIMUMPS process by MEMSCAP was selected. The process uses four masks to define regions to be etched and the deposition of metals. A device layer thickness of **25** μm was selected. The design area was **9** mm by **9** mm, but laser subdicing was used to provide for smaller sections. The laser subdicing was used to help to mitigate potential issues of dies being damaged during other dicing methods such as with a dicing saw.

[0044] The location of mechanical anchors were moved to the edge of the design. The SOIMUMPS process uses a backside etch release process, so this modification was needed to be compatible. Active frequency control was implemented by adjusting the bias voltage applied for the tuning electrodes. To allow for active frequency compensation that does not degrade phase noise, shaped combs were introduced to the device layout. Softening shaped combs were selected with a modest value for dimensional constant.

Experimental Setup

[0045] FIG. **6** shows an example of the experimental setup **600**. The parts were sent out for wire bonding with a local vendor. The selected packages **604** are custom 32-pin ceramic chip carriers **606**. To adhere the die **602** to the chip carrier **604**, the non-conductive epoxy Loctite Ablestik 789-3 was. In the past, a cyanoacrylate-based epoxy was used. Due to challenges with carefully placing the cyanoacrylate-based epoxy, the Loctite epoxy was used. To cure the epoxy, the dies **602** were heated to 150°C . for 30 minutes. After this, the dies **602** were wire bonded using standard 1 mil Au wire. To enable direct measurements of gravity, a 3D printed leveling mount **608** was created to hold the interface electronics board. The interface electronics board held the chip carrier that the die was bonded to. Using a simple bubble level as a reference, three screws equally spaced around the edge of the base can be adjusted to provide a consist means to directly measure local gravity.

[0046] The device discussed in this work requires a vacuum to function properly. A custom acrylic vacuum chamber **610** with a high number feedthroughs and a movable base was made by Sanatron. While the system was designed to reach a minimum pressure of 10 mTorr, it typically reaches above that value after an hour of pumping (about 15 mTorr).

[0047] An electronic subsystem was used to implement the accelerometer. The interface electronics board contained a socket to hold the packaged resonator for the accelerometer, active electronics to convert the charge output of the device to a voltage, and wiring to connect the resonator to other subsystems. To implement the needed single body multi-mode resonator **100**, a closed-loop system using the phase shifter and inverter/buffer were implemented. The phase shifter contained a digital potentiometer to enable automated adjustment of the phase shift. To control the amplitude of the single body multi-mode resonator **100**, an amplitude control circuit (AGC) was added. This circuit adjusts the supply limits of the inverter/buffer to control the amplitude of the single body multi-mode resonator **100** to a constant value. To address state-state error, a proportional-integral (PI) controller scheme was selected. The displacement of the single body multi-mode resonator **100** is digitized using a STEMLab 125-14 from Red Pitaya. Frequency detection, signal demodulation, and digital control loops were implemented with this device. For active frequency control, one of the digital to analog converters (DACs) on the STEMLab 125-14 was connected to an offset amplifier to level shift the output of the DAC to the one needed. A digital PI control loop was implemented to maintain the single body multi-mode resonator **100** at a constant frequency. To provide the voltages needed during the two different tuning phases, a function generator with a differential output was selected. To provide the needed reference for demodulation, the function generator was connected to the second input on the Red Pitaya.

[0048] Experimental Testing and Results

[0049] Using the experimental setup described herein, measurements from the accelerometer were taken. Results are shown in FIG. **7A**. As the device was tested using the leveling mount, the input

acceleration is assumed to be -1 g. To show the response of the different harmonics in the same figure, the signals were low-pass filtered and then the mean values were removed. All of the signals exhibit drift, however the drift rate of acceleration output, s1, is the greatest. This is clearer when the Allen deviation of these signals, shown in FIG. 7B, are calculated.

[0050] This device was tested while under active frequency control, so the frequency of the single body multi-mode resonator **100** does not drift, but the control voltage used to maintain a constant frequency provides an indirect means to measure this drift if the controller was disabled. Shown in FIG. 8A is a scatter plot of s1 and the control voltage. Using this signal to correct for drift, the Allen deviation is shown in FIG. 8B. Assuming that the output of s1 is 1 g, the results have been scaled. The advantage of this method is that the control voltage is less noisy, so the short-term stability is not compromised. The other is that the Allen deviation tends to decrease as the integration time is further increased.

[0051] As correlated noise processes are dependent on their prior state, the correlated noise was removed using an autoregressive filter of the form:

$$[00006] y_t = a_0 + a_1 t + \sum_{k=0}^p \alpha_k y_{t-k} + \epsilon_t$$

where the current state is modeled as a linear combination of prior values and an uncorrelated noise process. To mitigate the need to estimate a very high number of parameters for an autoregressive model, the signal calibrated using the control voltage was decimated to a slower sampling frequency. This decimated signal was then fit to an autoregressive model that automatically selects the model order based on optimizing the Akaike information criterion (AIC). The AIC of a given model is related to the log-likelihood of the model with a penalty for the number of parameters to balance issues related to over-fitting.

[0052] To demonstrate how well this method of using the control voltage to limit long-term drift and using an autoregressive model to limit short-term drift, the time series of the accelerometer output and the corrected versions are shown in FIG. 9A. The PSD of these signals are shown in FIG. 9B. With the signal filtered using the autoregressive model, the output of the sensor is dominated by uncorrelated noise. The Allen deviation of these signals are shown in FIG. 9C. On long time scale, the filtered accelerometer output does not exhibit drift. The bias instability, or the lowest point of the Allen deviation, is estimated to be about 55 ng.

[0053] As used herein, the term “about” is used to provide flexibility to a numerical range endpoint by providing that a given value may be “a little above” or “a little below” the endpoint. The degree of flexibility of this term can be dictated by the particular variable and would be within the knowledge of those skilled in the art to determine based on experience and the associated description herein.

[0054] As used herein, a plurality of items, structural elements, compositional elements, and/or materials may be presented in a common list for convenience. However, these lists should be construed as though each member of the list is individually identified as a separate and unique member. Thus, no individual member of a list should be construed as a de facto equivalent of any other member of the same list merely based on their presentation in a common group without indications to the contrary.

[0055] Unless otherwise stated, any feature described herein can be combined with any aspect or any other feature described herein.

[0056] Reference throughout the specification to “one example”, “another example”, “an example”, means that a particular element (e.g., feature, structure, and/or characteristic) described in connection with the example is included in at least one example described herein, and may or may not be present in other examples. In addition, the described elements for any example may be combined in any suitable manner in the various examples unless the context clearly dictates otherwise.

[0057] The ranges provided herein include the stated range and any value or sub-range within the stated range. For example, a range from about 2 μm to about 500 μm should be interpreted to

include not only the explicitly recited limits of from about 2 μm to about 500 μm, but also to include individual values, such as 100 μm, 200 μm, 350 μm, etc., and sub-ranges, such as from about 50 μm to about 150 μm, etc.

[0058] In describing and claiming the examples disclosed herein, the singular forms “a”, “an”, and “the” include plural referents unless the context clearly dictates otherwise.

Claims

1. A method for real-time frequency modulated bias calibration of short-term drift in a single-body multi-mode mechanical resonator accelerometer, comprising: measuring an extracted accelerometer output, wherein the extracted accelerometer output drifts over time, thereby causing an inaccurate extracted accelerometer output; applying an autoregressive filter to the extracted accelerometer output, thereby removing short-term noise; and calculating a corrected accelerometer output, thereby determining an acceleration.

2. The method of claim 1, wherein the autoregressive filter is applied using the equation (V):

$$y_t = \delta_{sub.0} + \delta_{sub.1}t + \sum_{k=0}^p \phi_{sub.k} y_{t-k} + \epsilon_{sub.t} \quad (V)$$
 where t is time, $\delta_{sub.0}$ is linear offset, $\delta_{sub.1}$ is linear trend with time, p is an autoregressive model order, $\phi_{sub.k}$ is an autoregressive model parameter, $y_{sub.t-k}$ is a model value at discrete time (t-k), k is a k-th term of the summation, and $\epsilon_{sub.t}$ is uncorrelated noise process at discrete time t.

3. The method of claim 1, wherein the autoregressive filter is fit to an autoregressive model that automatically selects the model order based on an Akaike information criterion.

4. The method of claim 1, wherein a computer processor, an application-specific integrated circuit, a field programmable gate array, or a microcontroller is used to record a measurement of the extracted output of the accelerometer, apply the autoregressive filter, and calculate the corrected accelerometer output.

5. The method of claim 4, wherein the extracted output of the accelerometer is obtained by measuring the an output of an interface electronic circuit that corresponds to a displacement of two or more spring-mass devices in an anti-phase mode, a differential charge produced by motion of two or more spring-mass devices in the anti-phase mode, or a combination thereof.

6. A method for real-time frequency modulated bias calibration of long-term drift in a single-body multi-mode mechanical resonator accelerometer, comprising: measuring an extracted accelerometer output, wherein the extracted accelerometer output drifts over time, thereby causing an inaccurate extracted accelerometer output; measuring the frequency of the single body multi-mode mechanical resonator accelerometer, thereby obtaining frequency data to extract a mean value; adjusting a voltage applied to a plurality of interdigitated shaped combs using the mean value and a PID controller, thereby forming a control voltage to maintain a constant mean frequency value; determining an amount of drift in the accelerometer using the control voltage; and calculating a corrected accelerometer output using the amount of drift, thereby determining an acceleration.

7. The method of claim 6, wherein a computer processor, an application-specific integrated circuit, a field programmable gate array, or a microcontroller is used to record a measurement output of the extracted accelerometer output, record a frequency measurement of the single body multi-mode resonator, adjust the voltage applied to the plurality of interdigitated shaped combs, determine the amount of drift in the accelerometer, and calculate the corrected accelerometer output.

8. A method for real-time frequency modulated bias calibration of short-term drift and long-term drift in a single-body multi-mode mechanical resonator accelerometer, comprising: measuring an extracted accelerometer output, wherein the extracted accelerometer output drifts over time, thereby causing an inaccurate extracted accelerometer output; performing one or both of: determining a corrected short-term accelerometer output by: i) applying an autoregressive filter to the extracted accelerometer output, thereby removing short-term noise; and ii) calculating the

corrected short-term accelerometer output, thereby determining a short-term acceleration; or determining a corrected long-term accelerometer output by: i) measuring the frequency of the single body multi-mode mechanical resonator accelerometer, thereby obtaining frequency data to extract a mean value; ii) adjusting a voltage applied to a plurality of interdigitated shaped combs using the mean value and a control algorithm, thereby forming a control voltage to maintain a constant mean frequency value; iii) determining an amount of drift in the accelerometer using the control voltage; and iv) calculating the corrected long-term accelerometer output, thereby determining a long-term acceleration.

9. The method of claim 8, wherein the determining the corrected short-term accelerometer output and the determining the corrected long-term accelerometer output occur simultaneously.

10. The method of claim 8, wherein determining the corrected long-term accelerometer output occurs before determining the corrected short-term accelerometer output.
